SOLAR-ASSIST ELECTRIC VEHICLE

ABSTRACT:

In the present situation energy crisis is an important unsolvable problem so we must find out some other ways to trust all sources such as solar energy, hydro power, tidal power, wind power etc. Using solar power to produce electricity is not the same as using solar to produce heat. Solar thermal principles are applied to produce hot fluids or air. Photovoltaic principles are used to produce electricity. In the present situation energy crisis is an important unsolvable problem so we must find out some other ways to trust all sources such as solar energy, hydro power, tidal power, wind power etc. Using solar power to produce electricity is not the same as using solar to produce heat. Solar thermal principles are applied to produce hot fluids or air. Photovoltaic principles are used to produce electricity.

INTRODUCTION:

Electric Vehicle (EV) technology is gaining ground and popularity rapidly. With depletion of oil reserves and a world characterized by smog, noise and all kinds of pollutants, governments and communities are awakening to the several benefits of EV technology. Zero emission vehicles are almost noiseless and can be charged at home or work, saving commuters endless queues at petrol stations. Charging at night when consumption is low, allows for efficient use of electricity. An electric car or battery electric car is an automobile that is propelled by one or more electric motors, using energy stored in batteries. Compared to internal combustion engine (ICE) vehicles, electric cars are quieter, and lower emissions overall. have no exhaust emissions.

BASIC PRINCIPLES:

When Light Hits the Cell When light, in the form of photons, hits our solar cell, its energy frees electron-hole pairs. Each photon with enough energy will normally free exactly one electron, and result in a free hole as well. If this happens close enough to the electric field, or if free electron and free hole happen to wander into its range of influence, the field will send the electron to the N side and the hole to the P side. This causes further disruption of electrical neutrality, and if we provide an external current path, electrons will flow through the path to their original side (the P side) to unite with holes that the electric field sent there, doing work for us along the way. The electron flow provides the current, and the cell's electric field causes a voltage. With both current and voltage, we have power, which is the product of the two.

OPERATION:

Solar vehicles are electric vehicles that use self contained solar cells to power themselves from sunlight. Solar vehicles typically contain a rechargeable battery to help regulate and store the energy from the solar cells and capture kinetic energy from the vehicle during braking.

A solar car is a solar vehicle for use on public roads or race tracks. Solar vehicles are electric vehicles that use self contained solar cells to power themselves from sunlight. Solar vehicles typically contain a rechargeable battery to help regulate and store the energy from the solar cells and capture kinetic energy from the vehicle during braking. Some solar cars can be plugged into external power sources to supplement the power of sunlight used to charge their battery.

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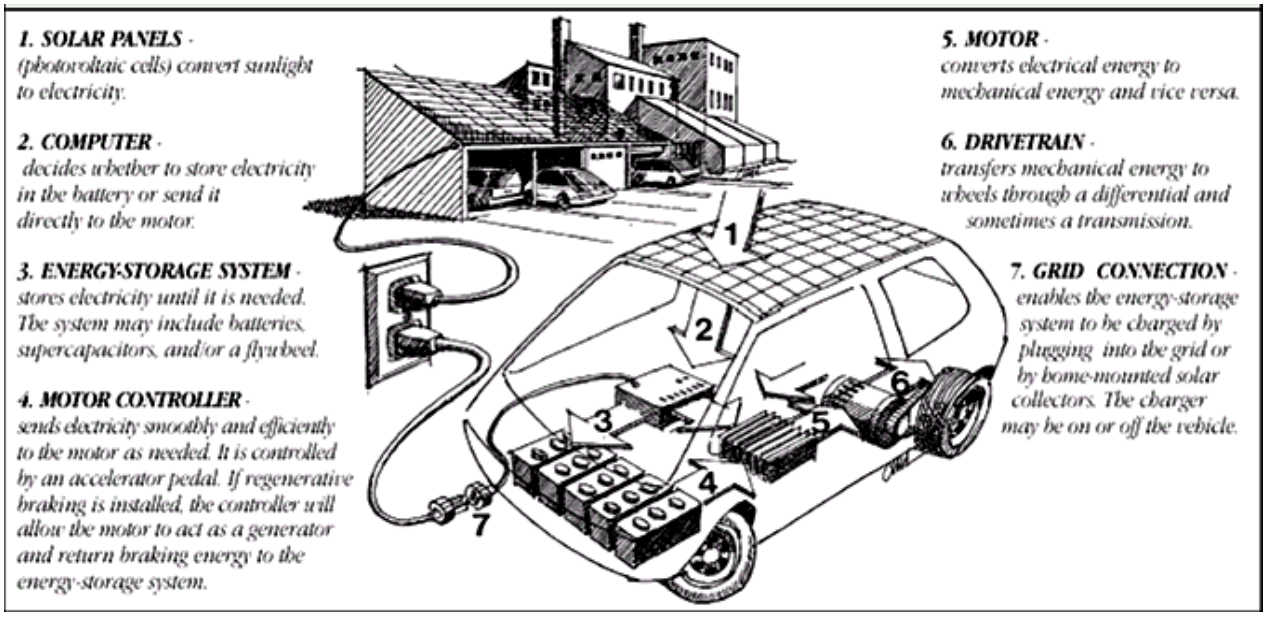

MERITS:

- Solar electricity systems are in many cases economically superior to conventional power systems.
- The operating costs are low because solar electricity needs very little maintenance, very few spare parts, and no fuel.
- The quality of the power supplied is excellent, and there are very few variations, if any
- Battery electric vehicle produces zero vehicular emissions.

DEMERITS:

- Limited energy stores at a time.
- Area required for collecting the sunlight is more. So shape of the vehicle is large compared to conventional vehicles.
- In cloudy days and in rainy seasons it has problems of collecting solar energy or to charge the battery.
- It has heavy weight.
- Cost of solar electric vehicle is very high.

CONCLUSION:

It is difficult to imagine a future without electric vehicles, which is probably the best alternative to conventional vehicles. Although technical and cost hurdles still remain it offers the best option for an environmentally safe personal transportation system.

FUTURE SCOPE:

Solar cars have been developing since the last twenty years and are powered by energy from the sun. Although they are not a practical or economic form of transportation at present, in the future they may play a part in reducing our reliance on burning fossil fuels such as petrol and diesel.

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